

VECTORS	
Term	Definition
Norm/Length	$\ v\ = \sqrt{\sum_{i=1}^n v_i^2}$ $\ v\ = \sqrt{v^T v}$ $\nabla \ v\ = \frac{1}{\ v\ } v^T$
Distance	$d(u, v) = \ u - v\ $ TODO: $\nabla d(u, v) = \frac{1}{d(u, v)} ((u - v)(u - v)) ???$ $\nabla_u d(u, v) = \frac{u - v}{\ u - v\ } = \frac{u - v}{d(u, v)}$ $\nabla_v d(u, v) = -\frac{u - v}{\ u - v\ } = \frac{v - u}{d(u, v)}$
Scalar product	$u \bullet v = u^T v = \sum_k u_k v_k$

SCALAR PRODUCT

MATRICES

Determinant:

$$\det(a) = a$$

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb$$

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$

Inverting:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad - cb} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Transposing:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{pmatrix}^T = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix}^T = (1 \ 4 \ 5)$$

Matrix multiplication:

$$\begin{pmatrix} 2 & 3 & 1 \\ 4 & -1 & 7 \end{pmatrix} \cdot \begin{pmatrix} 6 & 4 \\ 1 & 0 \\ 8 & 9 \end{pmatrix}$$

$$= \begin{pmatrix} (2 \cdot 6 + 3 \cdot 1 + 1 \cdot 8) & (2 \cdot 4 + 3 \cdot 0 + 1 \cdot 9) \\ (4 \cdot 6 + -1 \cdot 1 + 7 \cdot 8) & (4 \cdot 4 + -1 \cdot 0 + 7 \cdot 9) \end{pmatrix} = \begin{pmatrix} 23 & 17 \\ 79 & 79 \end{pmatrix}$$

INDICES

$$A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{m \times r}, A \cdot B \in \mathbb{R}^{n \times r}$$

$$(A \cdot B)_{ij} = \sum_{k=1}^m A_{ik} B_{kj} = \sum_k A_{ik} B_{kj}$$

$$((A \cdot B)^T)_{ij} = (A \cdot B)_{ji} = \sum_k A_{jk} B_{ik}$$

$$a^T \cdot B \cdot a = \sum_{ij} (a^T)_i B_{ij} a_j = \sum_{ij} a_i B_{ij} a_j$$

$$(B \cdot a)(C \cdot a) = \sum_i (B \cdot a)_i (C \cdot a)_i$$

KRONECKER DELTA

$$\delta_{ij} = (\vec{e}_i)_j = \begin{cases} 1, & i = j \\ 0, & \text{otherwise} \end{cases}$$

$$(E_{rst})_{ijk} = \delta_{ri} \delta_{sj} \delta_{tk}$$

BASIS VECTORS

$$\mathbb{R}^3$$

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

UNIT MATRICES

$$\mathbb{R}^2$$

$$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

AFFINE TRANSFORMATIONS

$$A_{a,M}(x) = a + M \cdot x$$

$$L_A \begin{pmatrix} \vec{0} \end{pmatrix} = \vec{0}$$

DERIVATIVES

$$\begin{array}{ll} 1 & \rightarrow 0 \\ x^2 & \rightarrow 2x \\ \frac{1}{x} & \rightarrow -\frac{1}{x^2} \\ e^x & \rightarrow e^x \\ \ln(x) & \rightarrow \frac{1}{x} \text{ for } x > 0 \\ a^x & \rightarrow \ln(a) \cdot a^x \text{ for } a > 0, a \neq 1 \\ \sin(x) & \rightarrow \cos(x) \\ \cos(x) & \rightarrow -\sin(x) \\ \tan(x) & \rightarrow 1 + \tan^2(x) = \frac{1}{\cos^2(x)} \\ \cot(x) & \rightarrow -\frac{1}{\sin^2(x)} \\ \arccos(x) & \rightarrow -\frac{1}{\sqrt{1-x^2}} \end{array}$$

$$\begin{array}{ll} x & \rightarrow 1 \\ x^n & \rightarrow nx^{n-1} \\ \sqrt{x} & \rightarrow \frac{1}{2\sqrt{x}} \\ e^{-x} & \rightarrow -e^{-x} \\ \ln(y \cdot x) & \rightarrow \frac{1}{x} \text{ for } x > 0 \\ \log_b(x) & \rightarrow \frac{1}{\ln(b)} \end{array}$$